$$\bar{H}_{O_{arms}} = \bar{r}_1 \times m_a \bar{v}_1 + \bar{r}_2 \times m_a \bar{v}_2$$

$$= 2 d \ell_a m_a \dot{\theta}_a \cos\theta_a \, \hat{j}$$

$$\bar{H}_{O_{legs}} = 2 d \ell_\ell m_\ell \dot{\theta}_\ell \cos\theta_\ell \, \hat{j}$$

$$\bar{H}_{O_{total}} = \bar{H}_{O_{arms}} + \bar{H}_{O_{legs}}$$

O is the CoM

If you're standing and swinging your arms, the GRF must counteract the varying angular momentum.

In practice, the trunk rotates slightly to the left and right during swinging of the arms in such a way as to make $H_O = 0$ at all times.

To avoid twisting of the trunk, total H_O should be zero

∴ Left arm must rotate in the same direction as the right leg & vice-versa

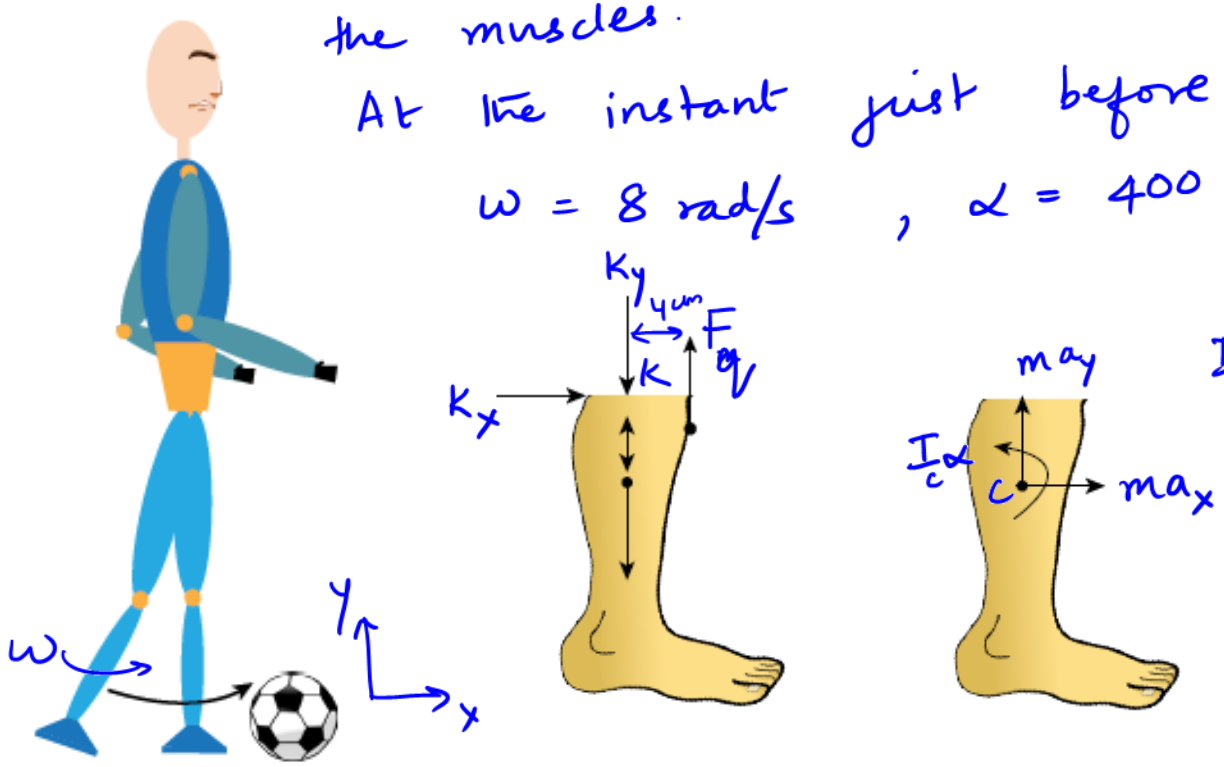
Also, lower limbs are longer & heavier than the arms

∴ arms must rotate faster to cancel out the twisting effect of the legs.

Since arms & legs must have the same time period of swing, to achieve zero angular momentum, we increase the amplitude of swing θ_a of the arms as well as the rate of rotation

Inverse dynamics – kicking a football

Inverse dynamics is used when external forces and kinematic data are known to compute the internal joint reactions & moments due to the muscles.



At the instant just before the foot strikes the ball,

$\omega = 8$ rad/s , $\alpha = 400$ rad/s^2, mass of lower leg = 7 kg

MoI about CoR @ knee $= 0.35$ kgm^2

Dist from knee to CoM of leg = 22 cm

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\Sigma \bar{M}_k = I_c \bar{\alpha} + \bar{r}_{c/k} \times m\bar{a}$$

$$F_{quad} = \underline{3500\ N}$$

Actual F_{quad} will be higher to counter the effect of the eccentric muscles!

Inverse dynamics – vertical jump

F_{calc} in the Achilles or calcaneal tendon

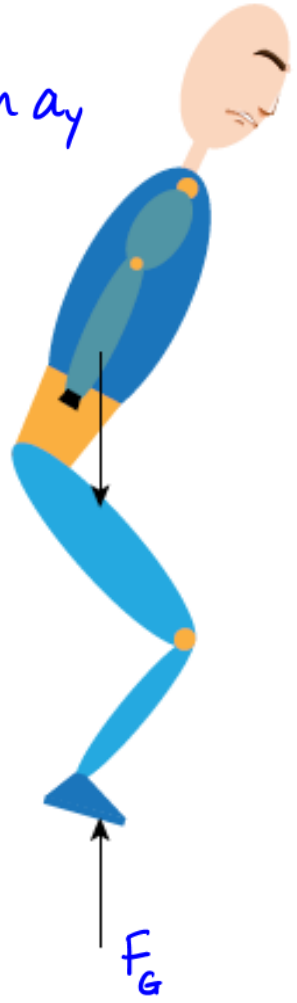
$m = 75$ kg

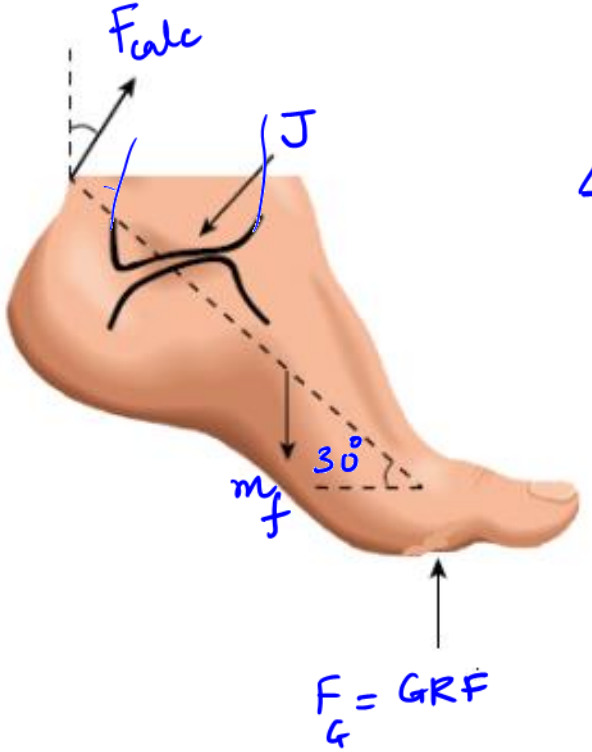
Vertical acc^n of the CoM at the instant of push-off $= 12\ m/s^2$

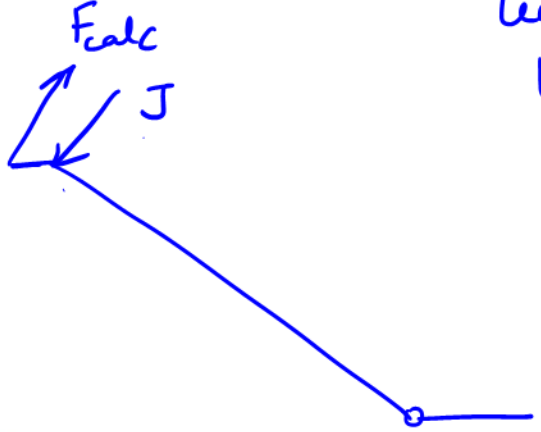
$2F_G - mg = ma_y$

$a_y = 12\ m/s^2 \uparrow$

$F_G = 825$ N







Length of foot = 27 cm

Length of toe region = 7 cm

Moment arm of the tendon = 4 cm

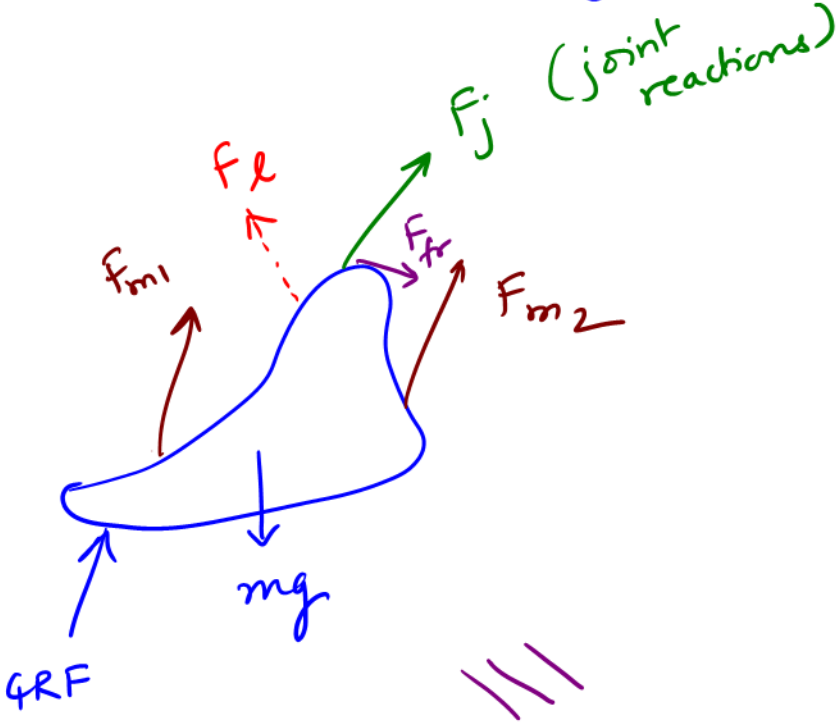
$\omega = -15$ rad/s

$\alpha = -150\ rad/s^2$

mass of foot = 1.5 kg = 15 N

Do a dynamic analysis & compare with a static analysis for the FBD of the foot

General inverse dynamic analysis procedure



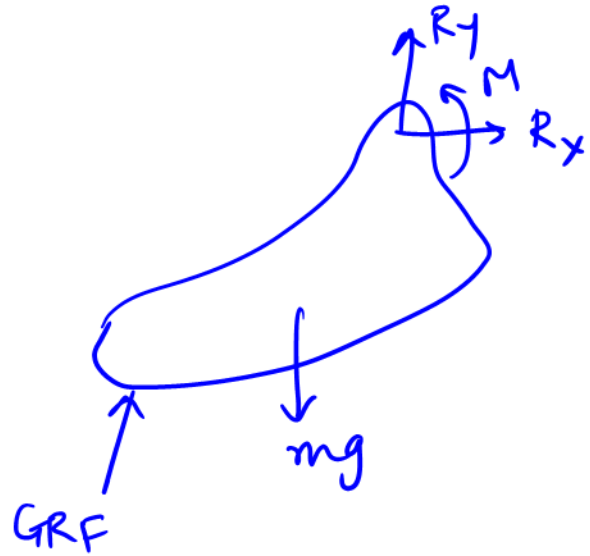
F_ℓ – ligamentous forces

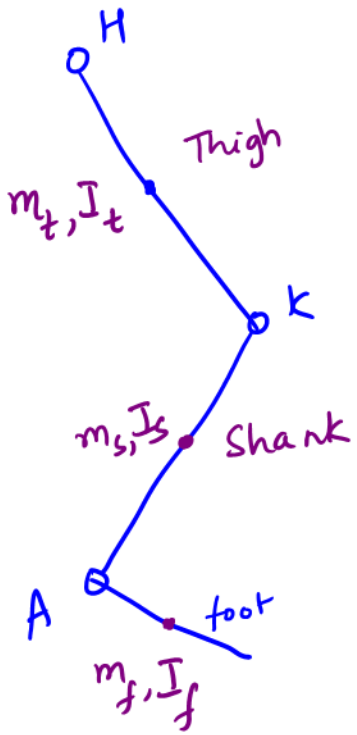
F_{fr} – joint friction

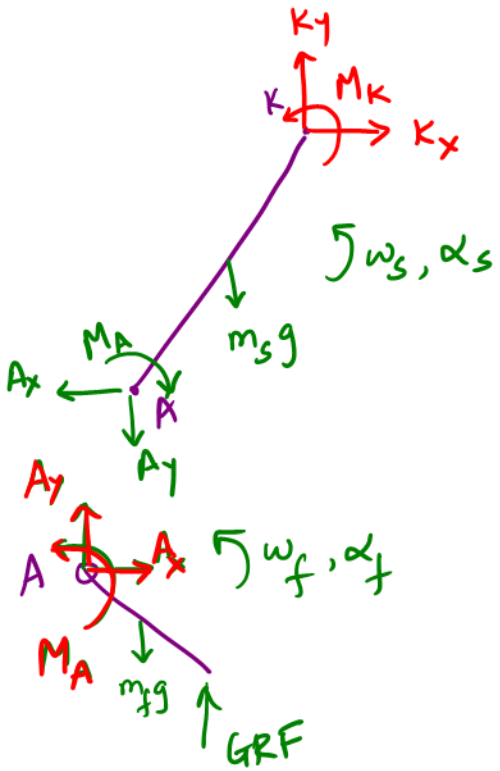
F_J – joint reaction

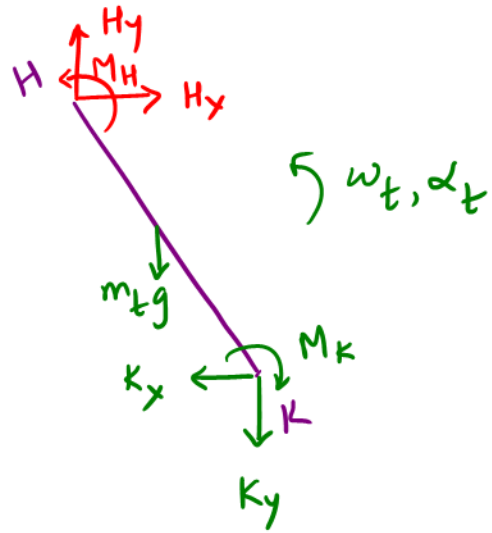
F_m – muscle forces

GRF – ground reaction force









GRF – from whole body analysis or force plate measurements

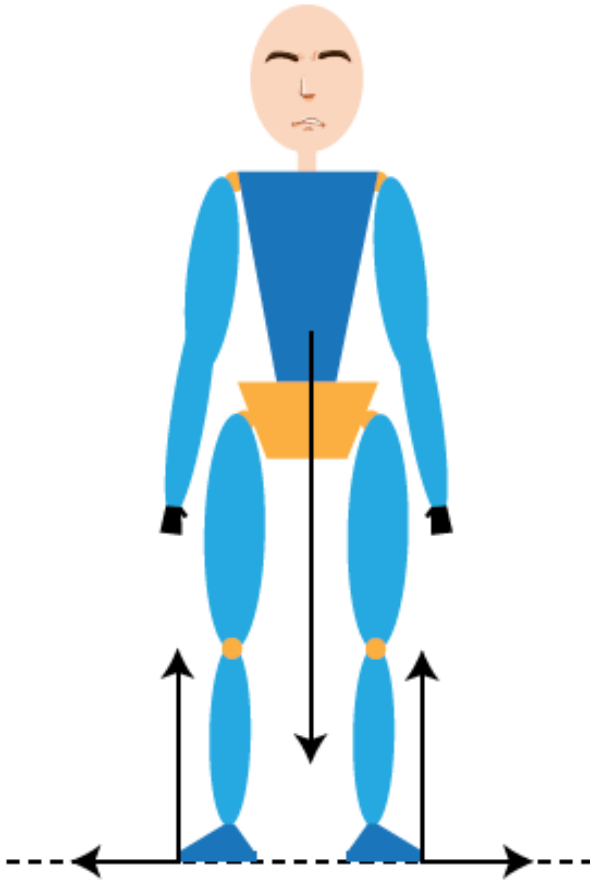
ω, α → motion capture system

direction of analysis

Starting from the lowest segment, all the unknowns are calculated step by step

For each segment $\sum F_x = ma_x$, $\sum F_y = ma_y$, $\sum M_{com} = I_{com}\ddot{\theta}$

Biomechanics of Balance



Body will remain in balance if CoM remains vertically above and within a certain prescribed area on the ground called as the base of support

